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**A Multi-Modal Markov-Chain Passenger-Flow Model for the BKK
Budapest GTFS Network: Entropy Maximisation, Kolmogorov
Equations, and Stochastic Simulation**

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Abstract

Research question. Which stops hold the Budapest public transport network together, and can this be answered from the published timetable alone, without fare-collection data? The paper asks how far a first-principles stochastic model built only from the BKK GTFS feed can describe passenger redistribution and identify critical stops.

Method. The timetable is turned into three stacked stochastic layers: a discrete-time Markov chain, a continuous-time chain whose generator combines Wilson's entropy-maximising destination kernel with scheduled departure frequencies, and exact Doob–Gillespie realisations with a τ -leaping approximation. Initial stop populations come from a prior anchored to published system ridership. Criticality is measured by node removal, combining stationary passenger throughput with the increase of Kemeny's constant and the loss of network efficiency, using only the components that are well defined for each chain. The same failure analysis is then inverted into a link-augmentation problem: new metro and tram links are chosen greedily to minimise the average and worst-case demand-weighted accessibility loss under single-station failures, using an exact shortest-path update.

Own results. On 5 177 stops the mean field agrees with exact simulation within 1% (metro) and with τ -leaping within 2.4% (full network). The metro chain is periodic, so spectral-gap measures are undefined. Criticality is concentrated in Deák Ferenc tér, Kálvin tér and Keleti pályaudvar; removing Deák alone cuts metro efficiency by 49%. Three short metro links that bypass Deák Ferenc tér and Kálvin tér (Rákóczi tér–Arany János utca, Fővám tér–Opera, Fővám tér–Vörösmarty tér) cut the worst-case loss from 44% to 19%, about twice the robustness gain of the planned M5 inner section, while pure line extensions add no redundancy at all.

Keywords: stochastic transit assignment, entropy maximisation, continuous-time Markov chain, Gillespie algorithm, GTFS, network resilience, Kemeny constant, network design

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I. Introduction

I.1. Motivation and Problem Setting

The Budapest metropolitan public transit network operated by Budapesti Közlekedési Központ (BKK) is one of Europe’s denser urban transit systems, serving on the order of 4×10^6 passenger journeys per weekday across six modal families (metro, bus, tram, trolleybus, suburban railway, and ferry). Despite this scale, spatially resolved passenger flow—the number of passengers at each stop at each point in time—is not directly observable from the publicly available schedule data alone.

The practical stakes are high. Spatial passenger density determines platform crowding risk, vehicle overcrowding, intermodal transfer delays, and the propagation of service disruptions. A principled model for passenger flow dynamics, even an imperfect one grounded in well-understood probabilistic machinery, enables:

1. **Network resilience assessment:** identifying stops and links whose failure disproportionately degrades service.
2. **Prior initialisation:** providing starting distributions for data-assimilation procedures once partial measurements (APC counters, AFC tap-in records) become available.
3. **Scenario analysis:** evaluating the system-level impact of new routes, closures, or timetable changes without requiring demand surveys.

I.2. Related Work

Entropy-based transport modelling. Wilson’s (Wilson, 1969) doubly-constrained gravity model remains the theoretical backbone of spatial interaction modelling. Its maximum-entropy derivation has been formalised and extended by Erlander and Stewart (Erlander and Stewart, 1990) and connected to random utility models by Anas (Anas, 1983). Recent work has embedded entropy transport models within the Wasserstein optimal-transport framework (Peyré and Cuturi, 2019).

Stochastic transit assignment. The logit-based stochastic user equilibrium (SUE) model of Dial (Dial, 1971) and the frequency-based assignment of Spiess and Florian (Spiess and Florian, 1989) are the industry standards. Dynamic stochastic assignment extending to time-varying supply is treated in Yildirimoglu and Kim (Yildirimoglu and Geroliminis, 2018) and the survey of Bliemer et al. (Bliemer et al., 2003). None of these papers couple the assignment to a CTMC generator or uses Gillespie simulation.

CTMC models for transit. Markov-chain models have been applied to pedestrian flow (Helbing et al., 2000), station waiting queues (Jaiswal, 1961), and subway dwell times (Zhang et al., 2019),

but full-network CTMC formulations over GTFS-structured schedules are uncommon. The closest precedent is the queueing-network approach of Heidergott et al. (Heidergott et al., 2010).

Gillespie simulation beyond biochemistry. Although Gillespie’s SSA (Gillespie, 1977) was developed for chemical reaction networks, it applies to any CTMC. Applications in ecology (Wilkinson, 2018), epidemiology (Allen, 2008), and traffic (Maerivoet and De Moor, 2005) demonstrate its scope.

Network resilience. The node-removal framework we adopt draws on Latora and Marchiori (Latora and Marchiori, 2001), Kemeny and Snell (Kemeny and Snell, 1976), and the transit-specific vulnerability analysis of Cats and Jenelius (Cats and Jenelius, 2014).

I.3. Contributions and Scope

- C1. CTMC generator from GTFS.** We show precisely how the Wilson entropy solution translates into a CTMC generator whose units are dimensionally consistent and whose entries are computable from `stop_times.txt`.
- C2. Regime analysis.** We compare the KFE mean-field trajectory with exact SSA and τ -leap ensembles on the real network and characterise when each layer is the appropriate model.
- C3. Sparse-graph formulation.** The full $n \times n$ cost matrix is never formed; all operations exploit the $\ll 1\%$ connectivity of the real GTFS graph.
- C4. Hierarchical demand prior.** A calibrated Poisson log-linear prior provides plausible initial passenger counts from freely available covariates.
- C5. Resilience ranking.** Three complementary vulnerability metrics (Kemeny’s constant, spectral gap, network efficiency) are combined with stationary passenger throughput into a stop criticality score, with reducible chains handled explicitly.
- C6. Robustness-driven augmentation.** The node-removal analysis is inverted into a link-augmentation problem. An exact shortest-path update makes greedy search over all candidate links cheap, three structural propositions explain which links can and cannot help, and the suggested links are compared with the projects planned for Budapest.

I.4. Paper Organisation

Section II formalises the network graph. Section III derives the entropy-maximising flow and transition probabilities. Section IV develops the CTMC generator and the Kolmogorov Forward Equation. Section V presents the hierarchical demand prior. Section VI covers the Gillespie SSA and τ -leaping. Section VII gives a computational complexity analysis. Section VIII describes the

Python implementation. Section IX reports empirical results on the full BKK network. Section X develops the resilience analysis. Section XI uses it to suggest new links for a more robust network. Section XII discusses limitations and future work. Section XIII concludes.

II. Network Graph and GTFS Data Model

II.1. GTFS Inventory

GTFS File	Key Fields Used	Role in Model
stops.txt	stop_id, stop_lat, stop_lon	Vertex set S , $ S = n$
routes.txt	route_id, route_type	Mode classification v
trips.txt	trip_id, route_id, service_id	Trip–route link
stop_times.txt	arrival_time, departure_time	Scheduled travel times $\bar{\tau}_{ij}^v$
shapes.txt	shape_pt_lat, shape_pt_lon	Route distance d_{ij}^v
calendar.txt	monday...friday	Weekday service filter
transfers.txt	from_stop_id, min_transfer_time	Transfer penalty δ_i

Table 1: BKK GTFS files used in this work.

Source: BKK GTFS Schedule feed (<https://opendata.bkk.hu>, CC0-1.0)

II.2. Multi-Modal Stop Graph

Definition II.1 (Multi-modal stop graph). Let $S = \{1, \dots, n\}$ denote the set of GTFS stops and $\mathcal{V} = \{1, \dots, m\}$ the set of vehicle modes. For each mode $v \in \mathcal{V}$, define the modal subgraph $G^v = (S^v, E^v)$, where $S^v \subseteq S$ is the set of stops served by mode v and

$$E^v = \{(i, j) \in S^v \times S^v : \exists \text{ a trip of mode } v \text{ visiting } i \text{ immediately before } j\}.$$

The multi-modal graph is the union $G = (S, \bigcup_v E^v)$.

Remark II.2 (Sparsity). On the BKK network $|E^v| \ll |S^v|^2$ for all modes. The bus sub-graph has a fill ratio $|E^v|/|S^v|^2$ of about 0.03% (Table 7). All matrices are stored in CSR format; the full $n \times n$ dense matrix is never materialised.

Definition II.3 (Neighbourhood sets). $\mathcal{N}^v(i) = \{j \in S^v : (i, j) \in E^v\}$, $\mathcal{N}_{\text{in}}^v(i) = \{j \in S^v : (j, i) \in E^v\}$.

v	Mode	rt	Speed (km/h)	Cap. (pax)	κ^v	Example Lines
1	Bus	3	18–25	80–180	1.00	7, 8E, 100E, 200E
2	Tram	0	15–22	200–350	0.70	2, 4, 6, 47–49
3	Trolleybus	800	15–20	100–170	0.85	70, 72, 74, 76
4	Metro	1	30–40	800–1000	0.40	M1, M2, M3, M4
5	Suburban rly	2	25–40	400–600	0.55	H5, H6, H7, H8
6	Ferry	4	12–18	100	1.20	D11, D12, D14

Table 2: BKK vehicle modes. κ^v is the mode-specific comfort weight; `rt` is the `route_type` code recognised by the framework.

Source: BKK extended GTFS route_type codes

II.3. BKK Mode Classification

II.4. Admissible Transitions and Disconnected Pairs

A transition (i, j) on mode v is admissible iff $(i, j) \in E^v$. We assign a finite cost only to admissible transitions; all others receive $C_{ij}^v = +\infty$, yielding $\exp(-\lambda^v C_{ij}^v) = 0$ and zero probability/rate. This handles graph sparsity exactly: we do not approximate disconnected pairs with a large finite penalty C_∞ , which would inject spuriously small but non-zero probabilities across the entire network.

Remark II.4 (Self-loops). The entropy-maximisation formulation excludes self-loops ($i \neq j$ in the constraint set); accordingly C_{ii}^v plays no role. The correct correspondence is that P^v as defined in Definition III.5 is the *embedded jump chain* of the CTMC, not the Δt -uniformisation.

II.5. Station-Level Aggregation of Metro Platforms

The BKK feed represents each metro station by two or more platform-level `stop_id` values, one per travel direction, linked to a common `parent_station`. Built naively, the metro graph therefore consists of one-directional paths: every trip enters a platform and leaves from the same platform in the same direction, so no platform can be reached again once left. Such a graph is a directed acyclic graph, it has no non-trivial strongly connected component, and the corresponding chain has no stationary distribution with support on the metro network.

We therefore define a node map $\sigma : S \rightarrow \tilde{S}$,

$$\sigma(s) = \begin{cases} \text{parent_station}(s), & \text{if defined,} \\ s, & \text{otherwise,} \end{cases} \quad (1)$$

and build the modal graphs on $\tilde{S}$. Edges with $\sigma(i) = \sigma(j)$ are dropped, travel times and distances of parallel edges are aggregated by the median, and the departure hazard of a station is computed from the number of distinct peak trips departing from any of its platforms. In the feed used here

537 stops.txt records have a parent station (platforms, entrances and generic nodes). Among the stops actually served, the 103 metro platforms collapse into 48 stations, and the resulting metro graph (48 nodes, 96 directed edges) is strongly connected, so its chain is irreducible. Only 26 served surface platforms (22 bus, 4 tram) have a parent station; all other surface stops keep their platform-level identifiers.

III. Entropy Maximisation and the Softmax Flow

III.1. Generalised Cost Matrix

Definition III.1 (Generalised cost). For admissible pair $(i, j) \in E^v$,

$$C_{ij}^v = \kappa^v \bar{\tau}_{ij}^v + \gamma d_{ij}^v + \delta \mathbf{1}_{\{i \in \mathcal{T}\}} t_{\text{tf}}(i), \quad (2)$$

where $\bar{\tau}_{ij}^v$ [s] is the median scheduled travel time, $\kappa^v > 0$ is the mode-specific comfort weight, d_{ij}^v [m] is the in-vehicle distance, γ [s m⁻¹] is the distance penalty, $t_{\text{tf}}(i)$ [s] is the minimum transfer time at stop i , and $\delta \in \{0, 1\}$ indicates a mode change. All terms are in seconds.

Remark III.2 (Dimensional consistency). C_{ij}^v [s] and λ^v [s⁻¹] make the exponent $\lambda^v C_{ij}^v$ dimensionless, consistent with the interpretation of λ^v as an inverse temperature governing cost sensitivity.

III.2. Entropy-Maximising Flow

Assumption III.3 (Separable origin constraint). Passengers are assigned a departure mode v before departure; within a mode, they choose destinations to maximise entropy subject to an expected-cost constraint.

Proposition III.4 (Wilson softmax flow). Under Assumption III.3, the flow $F_{ij}^v \geq 0$ maximising Shannon entropy $H = -\sum_{v,i,j} F_{ij}^v \ln F_{ij}^v$ subject to the origin constraint $\sum_j F_{ij}^v = N_i^v(0)$ and the cost constraint $\sum_{ij} F_{ij}^v C_{ij}^v \leq \bar{C}^v$ is given by

$$F_{ij}^v = N_i^v(0) \pi_{j|i}^v, \quad \pi_{j|i}^v = \frac{\exp(-\lambda^v C_{ij}^v)}{\sum_{k \in \mathcal{N}^v(i)} \exp(-\lambda^v C_{ik}^v)}, \quad (3)$$

where $\lambda^v \geq 0$ is the Lagrange multiplier, uniquely determined by the binding cost constraint.

Proof sketch. The Lagrangian yields $\ln F_{ij}^v = -1 + \mu_i^v - \lambda^v C_{ij}^v$, i.e. $F_{ij}^v \propto \exp(-\lambda^v C_{ij}^v)$. Row normalisation gives (3). Convexity of $-H$ ensures a unique maximiser. $\square$

Definition III.5 (DTMC transition matrix). The row-stochastic DTMC transition matrix for mode v is $P_{ij}^v = \pi_{j|i}^v$ ($i \neq j$), $P_{ii}^v = 0$ (no self-loop), $P_{ij}^v = 0$ for $(i, j) \notin E^v$.

III.3. DTMC Time Evolution

The population vector on mode v at step t evolves as

$$N^v(t+1) = N^v(t) P^v, \quad N_i(t) = \sum_{v=1}^m N_i^v(t). \quad (4)$$

IV. Continuous-Time Markov Chain and the Kolmogorov Forward Equation

IV.1. Motivation for the CTMC Formulation

The GTFS records precise departure timestamps at each stop. Rather than discretising the schedule into fixed steps Δt , we embed the passenger movement process in continuous time. Under the memoryless (Markov) approximation for inter-departure intervals—justified when vehicles run at roughly constant headway—the waiting time is exponentially distributed and the process is a CTMC.

Assumption IV.1 (Poisson arrivals). The departure process at stop i on mode v is approximated as a Poisson process with rate f_i^v [vehicles s^{-1}] estimated from the GTFS timetable.

IV.2. CTMC Generator

Definition IV.2 (Infinitesimal generator). For mode v , the generator $Q^v \in \mathbb{R}^{n \times n}$ is defined by

$$Q_{ij}^v = \begin{cases} q_{ij}^v \geq 0, & j \neq i, (i, j) \in E^v, \\ 0, & j \neq i, (i, j) \notin E^v, \\ -\sum_{k \in N^v(i)} q_{ik}^v, & j = i. \end{cases} \quad (5)$$

All off-diagonal entries are non-negative and all row sums vanish.

IV.3. Rates from Entropy Kernel and GTFS Frequencies

The off-diagonal rate decomposes as

$$q_{ij}^v = \underbrace{\mu_i^v}_{\text{departure intensity}} \cdot \underbrace{\pi_{j|i}^v}_{\text{Wilson kernel}}, \quad (i, j) \in E^v, j \neq i, \quad (6)$$

where the per-passenger departure hazard is the scheduled vehicle frequency

$$\mu_i^v = f_i^v \quad [\text{s}^{-1}]. \quad (7)$$

Here f_i^v [vehicles s^{-1}] is the peak-window departure frequency. This choice is dictated by dimensional consistency: every entry of a CTMC generator has units s^{-1} . Capacity $\bar{k}^v$ and load factor ρ^v describe supply and demand, but multiplying them into Q and then again by N_i in the KFE/SSA propensity would produce the invalid unit $\text{pax}^2 \text{s}^{-1}$.

IV.4. DTMC–CTMC Equivalence

Proposition IV.3 (Matrix exponential embedding). *The matrix exponential $P^v(\Delta t) := \exp(Q^v \Delta t)$ is stochastic for all $\Delta t \geq 0$; its first-order expansion recovers the Wilson DTMC entry structure with rescaled self-loop probabilities. The DTMC of Definition III.5 is the embedded jump chain of the CTMC, observed at transition moments stripped of holding times.*

IV.5. Kolmogorov Forward Equation

Theorem IV.4 (KFE). *Let $p^v(t) \in [0, 1]^n$ denote the normalised probability distribution of a single tagged passenger of mode v . Then*

$$\frac{dp^v(t)}{dt} = p^v(t) Q^v, \quad p^v(0) = p_0^v, \quad (8)$$

with unique solution $p^v(t) = p_0^v \exp(Q^v t)$.

By linearity and the mean-field (fluid-limit) approximation, the same equation holds for population counts:

$$\frac{dN^v(t)}{dt} = N^v(t) Q^v, \quad N^v(0) = N_0^v. \quad (9)$$

Conservation of mass follows from $Q^v \mathbf{1} = \mathbf{0}$.

IV.6. Stationarity, Ergodicity, and Existence of Stationary Distributions

Assumption IV.5 (Irreducibility per mode). For each mode v , the directed graph G^v is strongly connected.

Proposition IV.6 (Ergodicity). *Under Assumption IV.5, Q^v is irreducible and the CTMC is ergodic. A unique stationary distribution π^v with $\pi^v Q^v = \mathbf{0}$, $\|\pi^v\|_1 = 1$ exists, and $p^v(t) \rightarrow \pi^v$ exponentially fast.*

Remark IV.7 (Irreducibility in practice). The BKK graph is not strongly connected for every mode in isolation: the tram and bus graphs have 97 and 492 strongly connected components (`scipy.sparse.csgraph`). For such chains the stationary distribution is not unique; we use the uniform-start Cesàro average and treat irreducibility-dependent quantities explicitly (Section X.2).

Remark IV.8 (Time-homogeneity and GTFS schedules). A real GTFS schedule is time-varying; the time-homogeneous CTMC is valid only within a narrow service window (e.g., 07:00–09:00 peak hour) where f_i^v is approximately constant.

V. Prior Estimation of Initial Passenger Counts

V.1. Setting and Identifiability

In the absence of APC or AFC data, the initial count $N_i(0)$ is unobserved. We construct a prior $\hat{N}_i(0)$ from freely available covariates and calibrate its overall scale to BKK’s published aggregate ridership.

Assumption V.1 (Log-linear demand).

$$N_i(0) \sim \text{Poisson}(\lambda_i), \quad \log \lambda_i = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i + \xi_i, \quad \xi_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (10)$$

V.2. Feature Vector

Feature	Symbol	Source	Unit
Peak-hour service frequency	f_i	GTFS	trips h ⁻¹
Population in 400–800 m catchment	Pop_i	KSH/WorldPop	persons
POI count in 400 m catchment	POI_i	OpenStreetMap	count
Network betweenness centrality	B_i	GTFS graph	[0,1]
Interchange indicator	I_i	GTFS transfers	{0, 1}
Distance to Deák F. tér	D_i	Haversine	km

Table 3: Features used in the log-linear demand prior.

Source: own compilation from the transit-demand literature

V.3. Literature-Informed Prior Coefficients

Covariate	Symbol	μ_β	σ_β	Basis
$\log(1 + f_i)$	β_1	+0.70	0.10	Service elasticity (Schiewe and Lindner, 2022)
$\log(1 + \text{Pop}_i)$	β_2	+0.60	0.15	Population elasticity
$\log(1 + \text{POI}_i)$	β_3	+0.20	0.10	Activity generation
B_i	β_4	+0.30	0.15	Network centrality
I_i	β_5	+0.50	0.20	Transfer-hub effect
$\log(1 + D_i)$	β_6	-0.25	0.10	Distance decay

Table 4: Prior distributions $\mathcal{N}(\mu_\beta, \sigma_\beta^2)$ for log-linear coefficients.

Source: own compilation from the transit-demand literature

V.4. Calibration Anchor

BKK reports $N_{\text{day}} \approx 4 \times 10^6$ weekday boardings. With $\phi_{\text{peak}} \approx 0.11$,

$$N_{\text{peak}} = \phi_{\text{peak}} N_{\text{day}} \approx 440,000 \text{ passengers.} \quad (11)$$

The intercept β_0 is set so that $\sum_i \hat{N}_i(0) = N_{\text{peak}}$.

V.5. Diurnal Profile

$$\hat{N}_i(0; h) = 24 \phi(h) \hat{N}_i(0) \text{ with } \phi(07:00-09:00) \approx 0.11.$$

V.6. Three Estimators

E1 — Service proxy (GTFS only). $\hat{N}_i(0) \propto f_i \cdot \bar{k}^{v_i} \cdot \rho$. No external data needed; captures frequency variation but ignores land use.

E2 — Poisson log-linear (recommended). Equation (10) with priors from Table 4.

E3 — Full Bayesian (for future use with AFC/APC data). Full posterior inference via MCMC. Reduces to E2 when no observations are available.

V.7. Uncertainty Propagation

Monte Carlo propagation: draw $\beta^{(s)} \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$, compute $\hat{N}_i^{(s)}(0)$, run KFE or SSA, report empirical quantiles.

VI. Gillespie Stochastic Simulation Algorithm

VI.1. Reaction-Channel Representation

The propensity of channel $(v, i \rightarrow j)$ is

$$a_{ij}^v(t) = N_i^v(t) q_{ij}^v, \quad a_0(t) = \sum_v \sum_i \sum_{j \in N^v(i)} a_{ij}^v(t). \quad (12)$$

This is exact for $N_i^v(t) \in \mathbb{Z}_{\geq 0}$; the mean-field ODE (9) is recovered in the limit $N_i^v \rightarrow \infty$.

VI.2. Direct Method (Exact SSA)

Channel selection uses the $O(R)$ inverse-CDF method (`np.cumsum` and `np.searchsorted`). The Vose alias method (Vose, 1991) would give $O(1)$ selection but must be rebuilt after every event, because each event changes two propensities; for the channel counts considered here the simpler inverse-CDF method is sufficient.

Algorithm 1: Gillespie Direct Method for BKK Multi-Modal CTMC

Require: Sparse channels $\{(v, i, j, q_{ij}^v)\}$ (only $(i, j) \in E^v$); initial state $N(0)$; horizon T .

Ensure: Event list $\mathcal{E}$; final state $N(T)$.

1. $t \leftarrow 0, \quad \mathcal{E} \leftarrow \emptyset$
2. **while** $t < T$ **do**
 - (a) Compute propensities (vectorised): $a[r] \leftarrow N_i^v \cdot q_{ij}^v; \quad a_0 \leftarrow \sum_r a[r]$
 - (b) **if** $a_0 = 0$ **then break**
 - (c) Draw $\tau \leftarrow -\ln(u_1)/a_0, \quad u_1 \sim \mathcal{U}(0, 1)$
 - (d) **if** $t + \tau > T$ **then break**
 - (e) Select r^* via inverse-CDF on $\{a[r]/a_0\}$ using $u_2 \sim \mathcal{U}(0, 1)$
 - (f) $t \leftarrow t + \tau; \quad N_{i^*}^{v^*} \leftarrow N_{i^*}^{v^*} - 1; \quad N_{j^*}^{v^*} \leftarrow N_{j^*}^{v^*} + 1$
 - (g) Append (t, v^*, i^*, j^*) to $\mathcal{E}$
3. **return** $\mathcal{E}, N(T)$

Figure 1: Gillespie Direct Method for BKK Multi-Modal CTMC.

Source: own computation, `bkk/framework`

VI.3. τ -Leaping Approximation

When $N_i^v \gg 1$, the τ -leap (Gillespie, 2001) advances by a fixed step τ and draws

$$\Delta k_{ij}^v \sim \text{Poisson}(a_{ij}^v \tau), \quad (13)$$

then updates populations subject to an integer non-negativity guard. The implemented multinomial oversubscription rule preserves both mass and integer counts, while remaining an approximate leap rather than exact SSA.

Integer atomic outflow handling. If channels sharing origin i propose a total drain above N_i^v , independent serial clamping is order-dependent and proportional scaling creates fractional passengers. The implementation therefore redraws the N_i^v available passengers from a multinomial distribution with probabilities proportional to proposed counts:

$$\Delta k_{ij}^v \sim \text{Poisson}(a_{ij}^v \tau), \quad (14)$$

$$D_{vi} = \sum_{j \in \mathcal{N}^v(i)} \Delta k_{ij}^v, \quad (15)$$

$$(\Delta k_{ij}^v)_j \mid D_{vi} > N_i^v \sim \text{Multinomial}\left(N_i^v, (\Delta k_{ij}^v / D_{vi})_j\right). \quad (16)$$

Initial non-integer demand estimates are converted once by largest-remainder rounding, preserving the rounded grand total. Thereafter SSA and tau-leap states remain integer-valued.

Step-size control. The implementation uses the bounded propensity-ratio heuristic

$$\tau = \text{clip}\left(\varepsilon a_0 / \max_r a_r, 0.1 \tau_0, 10 \tau_0\right). \quad (17)$$

This is not the full Cao–Gillespie–Petzold method (Cao et al., 2006), which requires species-wise drift and variance bounds.

VI.4. Hybrid SSA–Fluid Decomposition

At stops with $N_i^v > N_{\text{thresh}} \approx 1000$, the coefficient of variation $\sim N_i^{-1/2} \ll 1$ and the fluid ODE (9) is accurate; at stops with $N_i^v \lesssim 10$ stochastic fluctuations dominate. We recommend a hybrid regime with coupling via boundary propensities at interface stops.

VII. Computational Complexity Analysis

VII.1. Graph-Structure Complexity

Let $n = |S|$, $m = |\mathcal{V}|$, $R = \sum_v |E^v|$. For the BKK feed used in this paper (Section IX): $n = 5,177$, $m = 4$, $R = 6,450$.

VII.2. One-Time Setup

Operation	Complexity	Notes
Parse GTFS files	$O(\text{trips} + \text{stop_times})$	pandas I/O
Build cost matrices C^v (CSR)	$O(R)$	Admissible pairs only
Build generator Q^v (CSR)	$O(R)$	
Strongly-connected components	$O(n + R)$	Tarjan
Betweenness centrality	$O(nm)$ sparse	NetworkX

Table 5: One-time setup complexity.

Source: own computation, bkk/framework

VII.3. Simulation Complexity

KFE (ODE solver): $O(R T_{\text{steps}})$.

Matrix exponential (Krylov): $O(R\ell)$ per time-point, $\ell = 30$ default.

Gillespie SSA: $O(a_0 T)$ events, each costing $O(R)$ with inverse-CDF selection, where $a_0 = \sum_{v,i} N_i^v f_i^v$. With the E1 prior anchored to $N_{\text{peak}} = 440,000$ passengers the full network has $a_0 \approx 5,600 \text{ s}^{-1}$, i.e. about 5.1×10^6 events in a 900 s window and $\sim 3 \times 10^{10}$ elementary operations. Exact SSA is therefore used on the metro sub-network ($a_0 \approx 814 \text{ s}^{-1}$), and τ -leaping on the full network.

τ -leaping: $O(R)$ Poisson draws per step (vectorised). With a fixed step $\tau = 1 \text{ s}$ and $T = 900 \text{ s}$ this is at most $900 \times R \approx 5.8 \times 10^6$ draws; the adaptive step-size rule needed about 91 steps on the full network.

Resilience: Naive dense perturbation is $O(Kn^3)$. ARPACK truncation reduces the spectral work, while node deletion still materialises a dense matrix; rank-one (Woodbury-type) updates (Hunter, 2014) are left for future work.

VII.4. Memory Requirements

The dense representation is three orders of magnitude larger than CSR; for the node-removal sweep, which forms many perturbed matrices, sparse storage is what keeps the analysis on a

Object	Dense	Sparse (CSR)
Bus generator Q^{bus}	~ 164 MB	~ 0.14 MB
All m generators	~ 167 MB	~ 0.16 MB
Simulation state $(N_i^v)_{v,i}$	~ 166 kB	—

Table 6: Memory footprint at BKK scale ($n = 5,177$, $R = 6,450$, $m = 4$).

Source: own computation, bkk/framework

single workstation.

VIII. Python Implementation

VIII.1. Software Stack

pandas / zipfile: GTFS ingestion. **numpy / scipy.sparse:** sparse matrices (CSR throughout). **scipy.sparse.linalg.expm_multiply:** Krylov action of $\exp(Qt)$. **scipy.integrate.solve_ivp:** ODE integration of (9). **networkx:** betweenness centrality, SCC decomposition. **overpy / osmnx:** OpenStreetMap POI queries. **rasterio:** WorldPop rasters for catchment population.

VIII.2. Core Implementation

Listing 1: Generator construction and KFE solve. The `GeneratorBuilder.build()` method groups edges by origin stop using `numpy.argsort`, applies the log-sum-exp stabilised softmax within each group (no Python-level for-loop over individual stop pairs), and assembles a CSR generator.

```

1 import numpy as np
2 from scipy.sparse import lil_matrix, csr_matrix
3 from scipy.integrate import solve_ivp
4
5 def build(mg, lam: float) -> csr_matrix:
6     """Build CTMC generator  $Q^v$  (CSR) without dense materialisation."""
7     i_arr, j_arr, cost_arr, mu_arr = (
8         mg.i_arr, mg.j_arr, mg.cost_arr, mg.mu_arr
9     )
10    n = mg.n_stops
11
12    # Sort edges by origin stop for numpy groupby
13    order = np.argsort(i_arr, kind='stable')
14    i_s, j_s, c_s = i_arr[order], j_arr[order], cost_arr[order]
15    q_vals = np.empty(len(order), dtype=np.float64)
16
17    boundaries = np.where(np.diff(i_s))[0] + 1
18    groups = np.split(np.arange(len(order)), boundaries)

```

```

19
20     for grp_idx in groups:
21         orig = i_s[grp_idx[0]]
22         costs = c_s[grp_idx]
23         # Log-sum-exp stabilised softmax:  $\pi^{v_{j|i}}$ 
24         log_w = -lam * costs
25         log_w -= log_w.max() # numerical stability shift
26         w = np.exp(log_w)
27         pi = w / w.sum()
28         #  $q^{v_{ij}} = \mu^{v_i} * \pi^{v_{j|i}}$ 
29         q_vals[grp_idx] = mu_arr[orig] * pi
30
31     # Assemble CSR generator Q
32     Q_lil = lil_matrix((n, n), dtype=np.float64)
33     for r in range(len(i_s)):
34         Q_lil[i_s[r], j_s[r]] = q_vals[r]
35     for i in range(n):
36         Q_lil[i, i] = -Q_lil.getrow(i).sum() # row sum = 0
37     return Q_lil.tocsr()
38
39
40 def kfe_solve_ode(Q: csr_matrix, N0: np.ndarray,
41                  T: float, t_eval: np.ndarray) -> np.ndarray:
42     """Solve mean-field ODE  $dN/dt = N Q$  via LSODA (sparse)."""
43     def rhs(t, N):
44         return Q.T.dot(N) # column form:  $d(N^T)/dt = Q^T N^T$ 
45     sol = solve_ivp(rhs, [0, T], N0.astype(float),
46                     t_eval=t_eval, method='LSODA',
47                     rtol=1e-8, atol=1e-10)
48     return sol.y.T # shape (len(t_eval), n)

```

Listing 1: Generator construction and KFE solve (bkk/generator.py, abridged).

Listing 2: Gillespie SSA — Direct Method with cumsum channel selection. The implementation matches Algorithm 1 exactly, using `np.searchsorted` on the cumulative propensity vector for $O(R)$ channel selection.

```

1 import numpy as np
2
3 def run_ssa(v_arr, i_arr, j_arr, q_arr, N, T, max_events, rng):
4     """
5     Exact Doob-Gillespie Direct Method ( $O(R)$  per event).
6
7     v_arr, i_arr, j_arr : int arrays shape (R,) -- channel indices
8     q_arr                : float array shape (R,) -- base rates  $q^{v_{ij}}$ 
9     N                    : float array shape (m, n) -- populations
10    T, max_events, rng    : horizon, safety cap, RNG
11    """

```

```

12     t = 0.0
13     events = []
14     for _ in range(max_events):
15         # Propensities: a[r] = N[v, i] * q[r]
16         a = N[v_arr, i_arr] * q_arr      # shape (R,)
17         a0 = a.sum()
18         if a0 <= 0.0:
19             break
20
21         # Waiting time ~ Exp(a0)
22         tau = -np.log(rng.random()) / a0
23         if t + tau >= T:
24             break
25         t += tau
26
27         # Channel selection via inverse-CDF (O(R))
28         u = rng.random() * a0
29         r = int(np.searchsorted(np.cumsum(a), u))
30         r = min(r, len(v_arr) - 1)
31
32         v, ii, jj = v_arr[r], i_arr[r], j_arr[r]
33         N[v, ii] = max(0.0, N[v, ii] - 1)
34         N[v, jj] += 1
35         events.append((t, v, ii, jj))
36
37     return events, N

```

Listing 2: Gillespie SSA (bkk/simulate.py, inner loop).

Listing 3: integer τ -leaping with atomic outflow handling. Oversubscribed origins use a multinomial allocation rather than fractional proportional scaling.

```

1  import numpy as np
2
3  def run_tauleap(v_arr, i_arr, j_arr, q_arr, N, T, tau, rng):
4      """
5      Integer tau-leaping with grouped multinomial oversubscription.
6      """
7      t = 0.0
8      while t < T:
9          tau_step = min(tau, T - t)
10
11         # Propensities and Poisson draws for all R channels
12         a = N[v_arr, i_arr] * q_arr      # (R,)
13         dk = rng.poisson(a * tau_step).astype(np.int64)
14
15         # Sum proposed outflow per (v, i)
16         total_out = np.zeros_like(N)      # (m, n)

```

```

17     np.add.at(total_out, (v_arr, i_arr), dk)
18
19     for v, i in np.argwhere(total_out > N):
20         channels = np.flatnonzero((v_arr == v) & (i_arr == i))
21         proposed = dk[channels]
22         dk[channels] = rng.multinomial(
23             int(N[v, i]), proposed / proposed.sum()
24         )
25
26     # Atomic update
27     np.add.at(N, (v_arr, i_arr), -dk)
28     np.add.at(N, (v_arr, j_arr), +dk)
29
30     t += tau_step
31     return N

```

Listing 3: τ -leaping with atomic outflow clamping (bkk/simulate.py).

Listing 4: E2 prior estimator.

```

1  import numpy as np
2
3  BETA_PRIOR = np.array([0.70, 0.60, 0.20, 0.30, 0.50, -0.25])
4
5  def compute_features(freq_i, pop_i, poi_i, B_i, I_i, D_i):
6      """Assemble log-transformed feature matrix X, shape (n, 6)."""
7      return np.column_stack([
8          np.log1p(freq_i),    # log(1 + f_i)
9          np.log1p(pop_i),     # log(1 + Pop_i)
10         np.log1p(poi_i),     # log(1 + POI_i)
11         B_i,                 # betweenness centrality [0, 1]
12         I_i,                 # interchange indicator {0, 1}
13         np.log1p(D_i),       # log(1 + distance to Deak ter)
14     ])
15
16  def estimate_N0(X: np.ndarray,
17                 beta: np.ndarray = BETA_PRIOR,
18                 N_peak: float = 440_000.0) -> np.ndarray:
19      """Anchored prior: sum_i N_hat_i(0) = N_peak."""
20      raw = np.exp(X @ beta)    # unnormalised intensities
21      return raw * (N_peak / raw.sum())

```

Listing 4: E2 prior estimator for $\hat{N}_i(0)$ (bkk/demand.py).

IX. Empirical Results on the BKK Network

IX.1. Data and Network Summary

The analysis uses the BKK GTFS Schedule feed released in June 2026 (<https://opendata.bkk.hu>, CC0-1.0). The archive contains 6 257 stop records, 386 routes, 247 616 trips and 678 054 shape points. The feed encodes service days through `calendar_dates.txt` only; selecting the service of a Monday retains 99 438 trips and 2 040 461 stop-time records. Edges are built from all weekday trips; departure hazards μ_i^v from the 07:00–09:00 peak window (245 866 departures). After station-level aggregation (Section II.5) the network has $n = 5,177$ nodes and $R = 6,450$ directed edges in four modes (Table 7).

Mode	rt	$ S^v $	$ E^v $	$ E^v / S^v $	Fill (%)	med $1/\mu_i^v$ (s)	med C_{ij}^v (s)
Tram	0	651	689	1.06	0.163	112	86
Metro	1	48	96	2.00	4.167	30	46
Bus	3	4 525	5 663	1.25	0.028	200	65
Ferry	4	2	2	1.00	50.000	900	218
Total	—	5 177	6 450	—	—	—	—

Table 7: Network summary by mode (weekday service, 07:00–09:00 peak window).

Fill = $|E^v|/|S^v|^2$. $1/\mu_i^v$ is the scheduled headway at stops with at least one peak departure. The total stop count is over distinct nodes; a node served by several modes is counted once.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0)

The graphs are extremely sparse: a surface stop has on average 1.06 (tram) or 1.25 (bus) scheduled successors, so for most stops the Wilson kernel has a single admissible destination and route choice happens only at branching stops. The metro graph is different in kind: a station has on average two successors (one per direction, more at interchanges, one at termini), the fill ratio is two orders of magnitude higher, and the median headway is 30 s, against 112.5 s for tram and 200 s for bus stops.

Mode coverage. The framework recognises trolleybus and suburban rail under the extended `route_type` codes 800 and 2 (Table 2). The current BKK feed codes them as 11 (16 trolleybus routes) and 109 (5 HÉV routes), so these two modes are not part of the network analysed below.

IX.2. Prior Estimate of Initial Counts

The results use the E1 service-proxy estimator, $\hat{N}_i(0) \propto \mu_i$, anchored to $N_{\text{peak}} = 440,000$; the E2 covariates (catchment population, POI counts) require external rasters and are left to the calibration stage. The prior is moderately concentrated: its Gini coefficient is 0.480, the 50 largest stops hold 5.6% of all passengers, and the maximum, 1 192 passengers, is at Deák Ferenc tér, where three metro lines meet (Table 8). The top of the list is formed by metro interchanges and by the bus platforms that feed them (Keleti pályaudvar, Blaha Lujza tér, Astoria).

Rk	Stop	stop_id	Mode	f_i (dep. h ⁻¹)	$\hat{N}_i(0)$
1	Deák Ferenc tér	CSF00954	Metro	333	1192
2	Kálvin tér	CS056227	Metro	242	868
3	Keleti pályaudvar	CS056233	Metro	241	863
4	Keleti pályaudvar M	F01159	Bus	200	714
5	Blaha Lujza tér M	F01166	Bus	161	576
6	Keleti pályaudvar M	F01314	Bus	160	571
7	Zugló vasútállomás	F02730	Bus	146	524
8	Zugló vasútállomás	F02725	Bus	140	503
9	Uránia	F01171	Bus	136	487
10	Astoria M	F01017	Bus	136	487

Table 8: Top-10 stops by the E1 prior $\hat{N}_i(0)$, anchored to $N_{\text{peak}} = 440,000$.

Metro platforms are aggregated to their parent station; surface stops keep one stop_id per platform, so a name can appear more than once.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0)

IX.3. KFE Dynamics

The population KFE (9) was solved over $T = 900$ s with $\lambda^v = 1/300$ s⁻¹ for all modes, using the Krylov action of $\exp(Q^{v\top}t)$. The largest relative change in total population over the horizon is 1.4×10^{-15} , i.e. passengers are conserved to machine precision, and all generators satisfy $\max_i |\sum_j Q_{ij}^v| < 10^{-17}$ and $\max_i |\sum_j P_{ij}^v - 1| < 3 \times 10^{-16}$.

Stop	$t = 0$	60 s	120 s	300 s	600 s	900 s
Deák Ferenc tér	1192	544	527	511	504	501
Kálvin tér	868	519	504	492	485	480
Keleti pályaudvar	863	415	405	388	370	360
Keleti pályaudvar M	714	230	237	132	63	40
Blaha Lujza tér M	576	315	223	124	56	27
Sum, initial top-50	24 517	19 749	18 054	16 464	15 590	15 142
Gini of $N(t)$	0.480	0.443	0.440	0.461	0.506	0.546

Table 9: Mean-field (KFE) passenger counts at the five stops with the largest initial count, full network, $\lambda^v = 1/300$ s⁻¹.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0)

Table 9 shows two time scales. At metro stations the population halves within the first minute (Deák Ferenc tér: 1 192 to 544) and then stays nearly constant: with a median headway of 30 s the metro chain mixes within a few headways, and what remains is its quasi-stationary level. The high-frequency bus platforms (Keleti pályaudvar M, Blaha Lujza tér M) lose passengers just as quickly but do not level off; they fall to 40 and 27 passengers by $t = 900$ s. The difference is structural: after aggregation the metro station graph is traversable in both directions, so passengers recirculate, whereas the bus graph is a collection of one-directional paths along which passengers are carried downstream towards the termini.

The Gini coefficient of $N(t)$ first falls, from 0.480 to 0.440 at $t = 120$ s, as the initial concentration

at high-frequency stops spreads along the lines, and then rises to 0.546 at $t = 900$ s. The rise is driven by *absorbing* nodes: 469 of the 5 226 mode-specific nodes either have no peak-hour departure ($\mu_i^v = 0$) or no outgoing edge, typically the arrival platforms of terminal stops. They hold 1.5% of the passengers at $t = 0$ and 19.2% at $t = 900$ s. The five largest stops at the horizon are all of this kind (for example Újpalota, Nyírpalota út: 360 to 3 473 passengers; Széll Kálmán tér M: 268 to 3 001). In an origin-constrained model without trip ends this is how alighting at the end of a line appears, and it is the main reason why the time-homogeneous KFE is meaningful only over a window of the order of the typical trip duration.

IX.4. SSA and τ -Leap Against the Mean Field

For a linear population CTMC the KFE is the exact equation of the first moment, so the comparison below is a test of the stochastic layers rather than of the model. At full scale an exact SSA run would require about 5×10^6 events per 900 s (Section VII), so the exact comparison is made on the metro sub-network, with the metro share of the prior (21 251 passengers), $T = 300$ s and $M = 20$ independent realisations.

Station	$\bar{N}^{\text{SSA}}$	N^{05}	N^{95}	KFE	CV (%)	$N^{-1/2}$ (%)
Bajcsy-Zsilinszky út	574.0	545	609	567.0	4.3	4.2
Pillangó utca	545.0	512	584	550.5	4.6	4.3
II. János Pál pápa tér	545.8	516	576	543.4	4.5	4.3
Puskás Ferenc Stadion	517.4	497	539	520.5	3.0	4.4
Blaha Lujza tér	521.6	495	548	519.5	3.7	4.4
Rákóczi tér	522.4	489	556	516.7	4.1	4.4

Table 10: Exact SSA ensemble ($M = 20$) against the KFE mean on the metro sub-network at $T = 300$ s, six largest stations.

CV: ensemble coefficient of variation; $N^{-1/2}$: the Poisson reference $1/\sqrt{N^{\text{KFE}}}$.
Source: own computation from the BKK GTFS Schedule feed (CC0-1.0)

Every realisation completed the full horizon (on average 223 571 events per run), so no run is censored by the event budget. Over all 48 stations the ensemble mean deviates from the KFE solution by 0.84% in relative ℓ^1 norm, and 43 of the 48 stations (90%) lie within two standard errors of the KFE value, consistent with the nominal 95% level for 48 stations. The ensemble coefficient of variation at the large stations is 3.0–4.6%, close to the reference value $N^{-1/2}$ (Table 10): at station level the fluctuations are essentially those of a Poisson count, and the mean field is an accurate description of every station with more than a few hundred passengers.

On the full network, ten τ -leap realisations (initial step $\tau_0 = 1$ s, $\varepsilon = 0.03$, on average 91 adaptive steps) conserve the integer population exactly and reproduce the KFE state at $t = 900$ s with a relative ℓ^1 deviation of 2.4%. For each of the five largest stops at the horizon the 5–95% ensemble band contains the KFE value. The deviation is larger than for exact SSA, as expected of an approximate leap method with multinomial redraws at nearly empty stops, but it remains

small compared with the uncertainty in the prior itself.

IX.5. Sensitivity to λ

Sweeping $\lambda \in [10^{-5}, 10^{-1}] \text{ s}^{-1}$ (25 log-spaced values) on the bus sub-network gives the following picture. (1) The mean row entropy of the transition matrix, $\bar{H}(P) = -\frac{1}{n} \sum_i \sum_j P_{ij} \ln P_{ij}$, decreases monotonically from 0.166 to 0.077 nats; its low level reflects the fact that most bus stops have a single successor. (2) The Gini coefficient of the (Cesàro) stationary distribution is essentially flat, between 0.53 and 0.54, for $\lambda \leq 5 \times 10^{-3} \text{ s}^{-1}$, and then rises steeply to 0.81 at $\lambda = 0.1 \text{ s}^{-1}$, when routing at branching stops becomes nearly deterministic. (3) The default $\lambda = 1/300 \text{ s}^{-1}$ ($\bar{H} = 0.163$, Gini 0.53) lies at the upper end of the flat regime, where the results are insensitive to the exact value of λ ; this matters because λ cannot be estimated without fare-collection data.

X. Network Resilience Analysis

X.1. Framework

Importance: the stationary departure flow of mode v at stop i ,

$$I_i^v = N_{\text{tot}}^v \pi_i^v \mu_i^v \quad [\text{pax s}^{-1}], \quad (18)$$

where N_{tot}^v is the prior population of mode v and π^v the stationary (for reducible chains: uniform-start Cesàro) distribution.

Vulnerability: Fractional degradation upon removing stop i^* (delete row/column i^* from P^v , renormalise):

$$V_{i^*}^{\text{Kem}} = \frac{K(P^{(-i^*)}) - K(P)}{|K(P)|}, \quad (19)$$

$$V_{i^*}^{\text{gap}} = \frac{g(P) - g(P^{(-i^*)})}{g(P)}, \quad (20)$$

$$V_{i^*}^{\text{eff}} = \frac{E(P) - E(P^{(-i^*)})}{E(P)}, \quad (21)$$

where, only for an irreducible chain, Kemeny's constant is $K(P) = \sum_{k=2}^n 1/(1 - \lambda_k(P))$ (Kemeny and Snell, 1976; Levene and Loizou, 2002), $g(P) = 1 - |\lambda_2(P)|$, and $E(P) = \frac{1}{n(n-1)} \sum_{i \neq j} 1/d_{ij}$ (Latora and Marchiori, 2001), with d_{ij} the hop distance in the support graph of P (for $n > 200$ estimated from the first 200 source nodes). V^{Kem} , V^{gap} and V^{eff} are truncated at zero. Dead-end rows are made absorbing. If node removal makes an irreducible chain reducible, $V^{\text{Kem}} = 1$ is an explicit fragmentation penalty rather than a fictitious Kemeny value. For reducible baselines, K

is reported as undefined and a uniform-start Cesàro stationary distribution is used.

Criticality:

$$C_i = \sqrt{\tilde{I}_i \tilde{V}_i}, \quad \tilde{V}_i = \frac{1}{|\mathcal{M}^v|} \sum_{c \in \mathcal{M}^v} \tilde{V}_i^c, \quad (22)$$

where every term is min-max normalised to $[0, 1]$ over the candidate set of mode v , and $\mathcal{M}^v \subseteq \{\text{Kem}, \text{gap}, \text{eff}\}$ is the set of components that are defined for the baseline chain of mode v : V^{Kem} requires an irreducible chain and V^{gap} a positive spectral gap $g(P) > 0$. Because of the normalisation, C_i ranks stops within a mode; it is not comparable across modes.

X.2. Well-Posedness of the Components on the BKK Chains

On the BKK network the spectral-gap component is undefined for every mode, for two different reasons. The tram and bus chains are reducible (97 and 492 strongly connected components, the largest with 234 and 4 009 nodes), so the eigenvalue 1 is multiple and $g(P) = 0$. The metro chain is irreducible but periodic with period 2: every line is traversed in both directions, so the station graph is bipartite and -1 is an eigenvalue of P^{metro} (the next largest moduli are 0.987). Again $g(P) = 1 - |\lambda_2| = 0$, and the relative change $(g - g^{(-i)})/g$ is a ratio of two numerically zero quantities. We therefore use $\mathcal{M}^{\text{metro}} = \{\text{Kem}, \text{eff}\}$ and $\mathcal{M}^{\text{tram}} = \mathcal{M}^{\text{bus}} = \{\text{eff}\}$. Kemeny's constant is well defined for the periodic metro chain ($K = 302.2$), since it only requires $\lambda_k \neq 1$ for $k \geq 2$. The two-node ferry chain is not ranked.

X.3. Computational Acceleration

Two speedups are used: (a) node removal is evaluated for the top- $K = 100$ candidates by importance in each mode (all 48 stations for the metro), and (b) for $n > 50$ only the largest-modulus eigenvalues are computed with ARPACK; for $n \leq 50$ the full spectrum is computed densely. Eigenvalues keep their sign and complex phase, since replacing λ by $|\text{Re } \lambda|$ would change Kemeny's sum; eigenvalues not computed by ARPACK are approximated by zero, each contributing 1 to K .

X.4. Numerical Results

X.4.1. Metro: criticality is concentrated in three interchanges

Table 11 ranks the metro stations. Three stations stand clearly apart: Deák Ferenc tér ($C_i = 1.000$), where M1, M2 and M3 meet, Kálvin tér (0.619; M3/M4) and Keleti pályaudvar (0.439; M2/M4). They carry the largest stationary departure flows (123, 60 and 43 pax s⁻¹, against about 15 pax s⁻¹ at an ordinary station), and removing Deák Ferenc tér alone reduces network efficiency by 49%, Kálvin tér by 32%. Below them the scores drop to 0.22–0.24 and are nearly flat; the order among

these stations is decided by the efficiency loss, which is largest for stations close to the city centre (Arany János utca, Corvin-negyed, Nyugati pályaudvar).

Rk	Station	I_i (pax s ⁻¹)	V^{Kem}	V^{eff}	C_i
1	Deák Ferenc tér	123.0	1.00	0.487	1.000
2	Kálvin tér	59.5	1.00	0.324	0.619
3	Keleti pályaudvar	43.3	1.00	0.096	0.439
4	Arany János utca	15.0	1.00	0.181	0.237
5	Corvin-negyed	14.9	1.00	0.158	0.232
6	Nyugati pályaudvar	15.0	1.00	0.148	0.231
7	Fővám tér	15.2	1.00	0.131	0.230
8	Lehel tér	15.1	1.00	0.122	0.227
9	Semmelweis Klinikák	14.7	1.00	0.128	0.224
10	Szent Gellért tér - Műgyetem	15.1	1.00	0.098	0.224
11	Dózsa György út	15.1	1.00	0.098	0.224
12	Göncz Árpád városközpont	15.3	1.00	0.076	0.221

Table 11: Metro: top-12 stations by criticality C_i (all 48 stations evaluated).

$V^{\text{Kem}} = 1$: removal disconnects the metro chain.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0)

The Kemeny component separates two groups. For 36 of the 48 stations the reduced chain is no longer irreducible: removing an intermediate station cuts its line into pieces that cannot reach each other, and the fragmentation penalty $V^{\text{Kem}} = 1$ applies. The remaining 12 stations fall into two groups. Seven are termini (for example Kelenföld vasútállomás, Örs vezér tere, Újpest-központ), whose removal shortens a line and does not increase K . The other five (Ferenciek tere, Astoria, Blaha Lujza tér, Rákóczi tér, II. János Pál pápa tér) lie on the inner cycle Deák Ferenc tér – Kálvin tér – Keleti pályaudvar formed by sections of M2, M3 and M4; removing one of them leaves an alternative route round the cycle, and K grows by only 6–26%. Kemeny’s constant thus acts mainly as a fragmentation indicator, which is what one expects of a network of four long lines with three transfer stations: only the inner cycle provides redundancy.

This is a structural statement about the timetable, not about observed crowding: no fare-collection data enter the model. The scheduled metro network routes a disproportionate share of its stochastic passenger flow through three interchange nodes, and this concentration is visible from the published GTFS feed alone.

X.4.2. Tram and bus

For the surface modes only the efficiency component is defined (Section X.2), and removing a single stop changes the efficiency of these large, sparse graphs very little: among the candidates the largest reduction is 3.1% for tram and 3.7% for bus stops, and at the high-throughput stops it is below 0.3% (Table 12). The tram ranking is led by Széll Kálmán tér M ($C_i = 0.281$), which combines by far the largest tram throughput (125 pax s⁻¹) with a measurable efficiency loss. The

following places are taken by stops of the northern Pest tram corridor (Újpest, Angyalföld), which have moderate throughput but the largest efficiency losses (2.8–3.1%). Since the efficiency of the tram and bus graphs is estimated from 200 source nodes, differences of this size should be read with caution. On the bus network the criticality scores are low throughout ($C_i \leq 0.05$); the list is headed by bus stops at metro stations (Móricz Zsigmond körtér, Kőbánya-Kispest) and at large bus terminals (Budaörsi lakótelep, Újpalota), i.e. by the feeder role of buses to the metro and to the outer districts. The small efficiency losses reflect the tree-like, largely one-directional structure of the surface graphs, in which a removed stop affects only paths through its own branch; a joint multi-modal chain with walking transfers would be needed to measure detours through other modes.

Rk	Stop	Mode	I_i (pax s ⁻¹)	V^{eff}	C_i
1	Széll Kálmán tér M	Tram	124.7	0.0025	0.281
2	Rákospalota-Újpest vasútállomás	Tram	1.9	0.0313	0.118
3	Lehel tér M	Tram	3.9	0.0143	0.117
4	Gyöngyösi utca	Tram	2.1	0.0283	0.116
5	Tél utca / Pozsonyi utca	Tram	2.1	0.0279	0.115
6	Angyalföld vasútállomás	Tram	2.0	0.0282	0.115
1	Móricz Zsigmond körtér M	Bus	142.7	0.0002	0.051
2	Kőbánya-Kispest M	Bus	300.4	0.0001	0.045
3	Budaörsi lakótelep	Bus	82.6	0.0001	0.033
4	Újpalota, Nyírpalota út	Bus	253.1	0.0000	0.031
5	Kőbánya-Kispest M	Bus	189.9	0.0001	0.030
6	Óbuda, Bogdáni út	Bus	86.3	0.0001	0.026

Table 12: Tram and bus: top-6 stops by criticality C_i among the 100 candidates with the largest stationary throughput.

The tram and bus chains are reducible, so C_i uses the efficiency component only.
Source: own computation from the BKK GTFS Schedule feed (CC0-1.0)

XI. Robustness-Driven Network Augmentation

Section X identifies the stations whose loss hurts the network most.¹ The natural next question for a planner is the inverse one: *which new links would reduce that damage the most?* This section turns the node-removal analysis into a link-augmentation problem for the metro and tram layers, solves it with an exact and fast shortest-path update, checks the greedy solution against exhaustive search, and compares the suggested links with the projects that are currently planned for Budapest.

¹[AUTHOR TO CONFIRM] In this section Claude (Anthropic) was used beyond the level stated in the AI declaration: for drafting text, the formulation of the propositions, the implementation in `scripts/paper_augment.py` and the literature search on planned projects. The author has checked the derivations and the results.

XI.1. Problem Statement

Planning layers. For each mode $v \in \{\text{metro}, \text{tram}\}$ we work on a station-level directed graph $G = (S, E)$ with arc weights equal to the generalised costs C_{ij} of Eq. (2). The metro layer is the 48-station graph of Section II.5. For the tram layer the directional platforms are merged into *stop areas*: platforms whose names agree after removing parenthesised qualifiers and which lie within 400 m of each other form one node. This turns the 651 tram platforms into 309 stop areas. The platform-level tram graph has 97 strongly connected components (Section X.2), but the stop-area graph is strongly connected: the reducibility of the tram chain is an artefact of one-directional platforms, not of the network.

Demand weights. Origin–destination pairs are weighted by the gravity form of the E1 prior of Section V,

$$\omega_{ij} = \frac{p_i p_j}{\sum_{k \neq l} p_k p_l}, \quad p_i = \frac{\hat{N}_i(0)}{\sum_k \hat{N}_k(0)}, \quad i \neq j, \quad (23)$$

so that pairs of busy stations count more than pairs of lightly used ones.

Candidate links. A candidate is an unordered pair $\{a, b\}$ of stations that are not adjacent in G . Adding it inserts the two arcs $a \rightarrow b$ and $b \rightarrow a$ with cost

$$c_{ab} = \hat{c}^v \ell_{ab}, \quad \hat{c}^v = \text{median}_{(i,j) \in E} \frac{C_{ij}}{\ell_{ij}}, \quad (24)$$

where ℓ_{ab} is the great-circle distance and $\hat{c}^v$ the median cost per metre of the existing links of the same mode (0.060 s m^{-1} for metro and 0.147 s m^{-1} for tram). The candidate set C^v contains the pairs with $150 \text{ m} \leq \ell_{ab} \leq \ell_{\max}^v$ whose current cost is at least 1.5 times the direct cost, $\min(D_{ab}, D_{ba}) \geq 1.5 c_{ab}$, so that the link is a real shortcut. We use $\ell_{\max}^{\text{metro}} = 3 \text{ km}$ and $\ell_{\max}^{\text{tram}} = 1.5 \text{ km}$. A new tram link may not cross the Danube, since that would need a new bridge. Metro links may cross it, as M2 and M4 already do. This gives $|C^{\text{metro}}| = 229$ and $|C^{\text{tram}}| = 579$.

Vulnerability of an augmented network. Let G_A denote G with the links of $A \subseteq C^v$ added, D^A its all-pairs cost matrix, and $G_A - k$ the same network after the failure of station k . With the demand-weighted efficiency restricted to the OD pairs that do not involve k ,

$$E_k(H) = \sum_{\substack{i \neq j \\ i, j \neq k}} \frac{\omega_{ij}}{D_{ij}^H}, \quad 1/\infty := 0, \quad (25)$$

the vulnerability of the augmented network to the loss of k is

$$V_k(A) = 1 - \frac{E_k(G_A - k)}{E_k(G_A)} \in [0, 1]. \quad (26)$$

$V_k(A)$ is the share of the (surviving) passengers' accessibility that is lost when k fails. The demand that starts or ends at k itself is excluded, because no new link can serve it. For $A = \emptyset$

and uniform ω this is closely related to the efficiency component V^{eff} of Section X, but uses costs instead of hop counts and excludes the pairs that start or end at k .

Objective. Following the choice of an average and a worst-case criterion, the robustness functional over a failure set $\mathcal{F} \subseteq S$ is

$$J(A) = \alpha \bar{V}(A) + (1 - \alpha) V_{\max}(A), \quad \bar{V}(A) = \frac{1}{|\mathcal{F}|} \sum_{k \in \mathcal{F}} V_k(A), \quad V_{\max}(A) = \max_{k \in \mathcal{F}} V_k(A), \quad (27)$$

with $\alpha = \frac{1}{2}$. The augmentation problem with a budget of B links is

$$A_B^* \in \arg \min_{A \subseteq C^v, |A| \leq B} J(A). \quad (28)$$

For the metro $\mathcal{F}$ contains all 48 stations. For the tram it contains the 100 stop areas with the largest prior demand, as in the node-removal sweep of Section X.

XI.2. Structural Results

The following statements make the problem tractable and explain the results.

Proposition XI.1 (Exact shortcut update). *Let D be the all-pairs shortest-path cost matrix of a directed graph with non-negative arc costs, and add the arcs $a \rightarrow b$ and $b \rightarrow a$ with cost $c \geq 0$. The new cost matrix is*

$$D'_{ij} = \min\{D_{ij}, D_{ia} + c + D_{bj}, D_{ib} + c + D_{aj}\}. \quad (29)$$

The same formula applied to $D^{(-k)}$, the cost matrix of $G - k$, gives the cost matrix of $G_{\{ab\}} - k$ for every $k \notin \{a, b\}$; for $k \in \{a, b\}$ the matrix is unchanged.

Proof. Every expression on the right is the cost of a walk from i to j in the new graph, so D'_{ij} is at most the minimum. Conversely, take a shortest i – j path in the new graph. With non-negative costs we may assume that it is simple, so it uses at most one of the two new arcs: using $a \rightarrow b$ and later $b \rightarrow a$ (or the reverse) would visit a twice. If it uses no new arc its cost is at least D_{ij} . If it uses $a \rightarrow b$, its parts before and after this arc are paths of the old graph from i to a and from b to j , so its cost is at least $D_{ia} + c + D_{bj}$; the case $b \rightarrow a$ is symmetric. When k has been removed, a link with an endpoint at k does not exist, and otherwise the same argument holds in $G - k$. $\square$

Applying (29) link by link gives the cost matrices for any set A . After a one-time computation of D and of $D^{(-k)}$ for all $k \in \mathcal{F}$ ($|\mathcal{F}| + 1$ runs of Dijkstra's algorithm), evaluating a candidate costs $O(|\mathcal{F}| n^2)$ elementary operations and requires no further shortest-path computation. In the implementation the update agrees with a full recomputation to 10^{-13} s.

Proposition XI.2 (Extensions do not add redundancy). *Let the augmentation attach a new subgraph H (for example a line extension with new stations) to G at a single station s . Then $D_{ij}^A = D_{ij}$ for all $i, j \in S$ and, with the demand weights of the existing stations, $V_k(A) = V_k(\emptyset)$ for all $k \in S$.*

Proof. A path between $i, j \in S$ that enters H has to enter and leave it through s , so it contains a closed walk through s . Removing that walk leaves a path in G that costs no more. Hence all costs within S , with or without the failure of any $k \in S$, are unchanged, and so are the efficiencies in (26). $\square$

The planned M3 extension to Rákospalota-Újpest (Telex, 2026), the through-running of M2 onto the Gödöllő HÉV (Beruházás.gov.hu, 2026) and a possible M4 extension to Bosnyák tér are extensions of this kind. They improve coverage, but by Proposition XI.2 they do nothing for the robustness of the existing network. If the new stations carry demand, the failure of the attachment station s also cuts them off, so s becomes *more* critical.

Proposition XI.3 (Hub-reinforcement trap). *If every link in A has an endpoint at k , then $V_k(A) \geq V_k(\emptyset)$.*

Proof. The links of A disappear together with k , so $G_A - k = G - k$ and the numerator of (26) is unchanged, while adding arcs can only shorten paths, so $E_k(G_A) \geq E_k(G)$. $\square$

New links into an already critical interchange raise the accessibility that is lost when that interchange fails, so they make it more critical. Robustness improves through links that *bypass* a hub, not through links that feed it.

Remark XI.4 (Hardness and the greedy heuristic). The closely related problem of adding B shortcut edges to minimise the average shortest-path distance is NP-hard (Meyerson and Tagiku, 2009); (28) adds failures and a worst-case term, and no exact polynomial algorithm is known for it. The reduction $J(\emptyset) - J(A)$ is neither monotone (by Proposition XI.3 a link can increase some V_k) nor submodular in general, since J contains a maximum. The $(1 - 1/e)$ guarantee of the greedy algorithm for monotone submodular maximisation (Nemhauser et al., 1978) therefore does not apply. We use the greedy algorithm and measure its gap empirically against exhaustive search for $B = 2$.

Remark XI.5 (Periodicity). The metro chain is periodic because its support graph is bipartite (Section X.2). A new link between two stations of the same colour class creates an odd cycle. Since every metro link is used in both directions, the chain also has cycles of length 2, so its period becomes $\gcd(2, \text{odd}) = 1$. Such a link therefore makes the chain aperiodic and the spectral gap $g(P)$ positive, which makes the component V^{gap} of Section X well defined again.

XI.3. Algorithm

Algorithm 2: Greedy robustness-driven augmentation

Require: layer G , costs C , weights ω , failure set $\mathcal{F}$, candidates C , budget B , committed links A_0 (empty, or a planned project).

Ensure: links $a_1, \dots, a_B$ and J after each step.

1. Compute D and $D^{(-k)}$, $k \in \mathcal{F}$, by Dijkstra; apply (29) for the links of A_0 . $A \leftarrow A_0$.
2. **for** $b = 1, \dots, B$ **do**
 - (a) **for each** $e \in C \setminus A$: update D and all $D^{(-k)}$ with e by (29), and evaluate $J(A \cup \{e\})$ by (26)–(27).
 - (b) $a_b \leftarrow \arg \min_e J(A \cup \{e\})$; $A \leftarrow A \cup \{a_b\}$.
3. **return** $a_1, \dots, a_B$

Figure 2: Greedy robustness-driven link augmentation.

Source: own computation, scripts/paper_augment.py

The cost is $O(B|C||\mathcal{F}|n^2)$: about 1.3×10^8 operations for the metro with $B = 5$, and about 2.7×10^{10} for the tram. For the tram the candidate list is therefore first reduced to the 100 best single links by $J(\{e\})$. The metro run, including the exhaustive check of all $\binom{229}{2} = 26\,106$ pairs, takes about two minutes on a laptop.

XI.4. Results: Metro

With the cost-based, demand-weighted definition the vulnerabilities of the current metro reproduce the ranking of Section X.4.1: the failure of Deák Ferenc tér removes 44.5% of the remaining accessibility and Kálvin tér 31.1%, while no other station exceeds 16%. The worst-case term of J is thus set by the two central interchanges.

Table 13 shows the greedy plan. The first three links, Rákóczi tér – Arany János utca (1.7 km), Fővám tér – Opera (1.8 km) and Fővám tér – Vörösmarty tér (1.3 km), form a ring of connections around Deák Ferenc tér and Kálvin tér. Each of them joins two lines whose only connection today runs through one of the two hubs (M3 and M4 meet only at Kálvin tér; M1 reaches M4 only through Deák Ferenc tér). Together they cut the worst-case loss from 44.5% to 18.8% and reduce J by 52.1%. The fourth and fifth links (Hősök tere – Dózsa György út on the M1/M3 axis and II. János Pál pápa tér – Népliget on the M3/M4 axis) mainly lower the average $\bar{V}$, because they give the outer sections of the lines a second connection. In line with Proposition XI.3, none of the chosen links ends at Deák Ferenc tér or Kálvin tér.

b	Added link	km	$\bar{V}$	$V_{\max}$	Worst failure	J	$-\Delta J$ (%)
0	<i>current network</i>	—	0.070	0.445	Deák Ferenc tér	0.257	—
1	Rákóczi tér – Arany János utca	1.73	0.067	0.328	Deák Ferenc tér	0.198	23.3
2	Fővám tér – Opera	1.77	0.060	0.208	Deák Ferenc tér	0.134	48.1
3	Fővám tér – Vörösmarty tér	1.26	0.058	0.188	Kálvin tér	0.123	52.1
4	Hősök tere – Dózsa György út	1.53	0.046	0.185	Kálvin tér	0.115	55.1
5	II. János Pál pápa tér – Népliget	2.73	0.038	0.170	Deák Ferenc tér	0.104	59.7

Table 13: Metro: greedy robustness-driven link additions. Each row adds one bidirectional link to all previous ones. $\bar{V}$ and $V_{\max}$ are the mean and maximum demand-weighted vulnerability over single-station failures, $J = \frac{1}{2}(\bar{V} + V_{\max})$, and the last column is the cumulative reduction of J .

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0), `scripts/paper_augment.py`

Greedy against exhaustive search. For $B = 2$ the exhaustive optimum (Fővám tér – Bajcsy-Zsilinszky út and Arany János utca – Bajcsy-Zsilinszky út) gives $J = 0.127$, the greedy pair $J = 0.134$. The greedy solution is within 5% of the optimum, and both optima build the same kind of bypass between the M1, M3 and M4 stations around the two hubs.

Sensitivity. The first three greedy links are identical for $\ell_{\max} = 2$ km, 3 km and 4 km, and for the pure worst-case objective ($\alpha = 0$). They depend on the weighting, however. With the pure average ($\alpha = 1$) the greedy algorithm connects the outer sections of the lines instead (Göncz Árpád városközpont – Széchenyi fürdő, Vörösmarty utca – Semmelweis Klinikák, Újbuda-központ – Vörösmarty tér), and with uniform demand weights it chooses Kálvin tér – Opera, Fővám tér – Déli pályaudvar and Hősök tere – Dózsa György út. The inner bypass ring is therefore the answer to the worst-case question. A planner who cares only about the average failure would build different links.

Metro network	$K(P)$	$g(P)$	periodic	fragmenting stations
current network	302.2	0.0000	yes	36
greedy, $b = 1$	286.3	0.0103	no	36
greedy, $b = 2$	262.9	0.0105	no	35
greedy, $b = 3$	259.1	0.0148	no	35
greedy, $b = 4$	220.2	0.0150	no	27
greedy, $b = 5$	193.2	0.0152	no	23
M5 inner section	237.4	0.0086	no	32
M5 inner section + 3 next links	201.7	0.0204	no	25

Table 14: Markov-chain diagnostics of the augmented metro chain ($\lambda = 1/300 \text{ s}^{-1}$). The last column counts the stations whose removal leaves a chain that is not irreducible.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0), `scripts/paper_augment.py`

Effect on the Markov chain. Table 14 evaluates the augmented networks with the same Wilson generator as in Section IV. The first greedy link already makes the metro chain aperiodic (Remark XI.5), so its spectral gap becomes positive. Kemeny’s constant falls from 302.2 to 193.2 after five links, i.e. a random passenger reaches a random destination in fewer steps. The number of stations whose loss disconnects the chain falls from 36 to 23.

XI.5. Results: Tram

The tram stop-area graph is much less vulnerable than the metro: the largest loss, at the Lehel utca / Róbert Károly körút stop area where line 1 crosses lines 12/14, is 8.9%, followed by Széll Kálmán tér (8.6%). The surface network already has many parallel routes, which matches the small efficiency losses of Section X.4.2.

b	Added link	km	$\bar{V}$	$V_{\max}$	Worst failure	J	$-\Delta J$ (%)
0	<i>current network</i>	—	0.024	0.089	Lehel utca / Róbert Károly körút	0.057	—
1	Vágány utca / Róbert Károly körút – Hun utca	0.47	0.024	0.086	Széll Kálmán tér M	0.055	3.5
2	Batthyány tér M+H – Déli pályaudvar M	1.25	0.023	0.080	Széll Kálmán tér M	0.052	8.6
3	Oktogon M – Lehel tér M	1.30	0.021	0.080	Széll Kálmán tér M	0.051	10.7
4	Blaha Lujza tér M – Kálvin tér M	1.07	0.020	0.080	Széll Kálmán tér M	0.050	12.3
5	Budafoki út / Dombóvári út – Budafoki út / Szerémi sor	0.47	0.019	0.079	Széll Kálmán tér M	0.049	13.1

Table 15: Tram: greedy robustness-driven link additions. Each row adds one bidirectional link to all previous ones. $\bar{V}$ and $V_{\max}$ are the mean and maximum demand-weighted vulnerability over single-station failures, $J = \frac{1}{2}(\bar{V} + V_{\max})$, and the last column is the cumulative reduction of J .

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0), `scripts/paper_augment.py`

The greedy tram plan (Table 15) consists of short gap closures. (1) A 0.5 km link Vágány utca / Róbert Károly körút – Hun utca joins line 1 to lines 12/14 away from their only crossing and removes the worst case. (2) Batthyány tér – Déli pályaudvar (1.2 km) connects the riverside lines 19/41 with lines 17, 56, 59 and 61 without passing through Széll Kálmán tér. (3) Oktogon – Lehel tér (1.3 km) joins the Grand Boulevard lines 4/6 to lines 12/14. (4) Blaha Lujza tér – Kálvin tér (1.1 km) joins lines 4/6 and 28/37 to lines 47/49 along the inner boulevard. (5) A 0.5 km gap on Budafoki út joins line 1 to line 4. It lies on the corridor of the Budafoki út tram line, which is currently being designed (BKK, 2026). Five links reduce J by 13.1%. The worst case moves to Széll Kálmán tér, whose loss the five links reduce only from 8.6% to 7.9%.

XI.6. Planned Projects and Suggested Next Links

Table 16 evaluates the projects that are currently planned or discussed with the same functional.

M5 inner section. The inner section of the north–south regional rail (M5) would run from Boráros tér through Kálvin tér, Astoria and Oktogon to Lehel tér (Wikipedia contributors, 2026). Represented by its three links between existing metro stations (the new station at Klauzál tér is not a node of the current graph), it lowers J by 27.8%. This is roughly half of what the greedy plan achieves with the same number of links (52.1%). The reason is Proposition XI.3 at work: the M5 passes through Kálvin tér, the second most critical station, and makes it the new worst case (32.0%). The M5 is still the best of all evaluated plans for the intact network: it raises

Layer	Project	Links	km	$\bar{V}$	$V_{\max}$	$-\Delta J$ (%)	greedy, same b (%)
Metro	M5 inner section (Kálvin tér–Astoria–Oktogon–Lehel tér)	3	3.2	0.052	0.320	27.8	52.1
Tram	Pesti fonódó (Deák tér–Nyugati tér–Lehel tér)	2	2.3	0.021	0.096	−2.7	8.6
Tram	Thököly út tram (Keleti pu.–Bosnyák tér), discussed	1	3.2	0.023	0.089	1.1	3.5

Table 16: Planned and discussed projects evaluated on the same robustness functional. The last column is the reduction achieved by the greedy plan with the same number of links.

Pure extensions (M3 to Rákospalota–Újpest, M2–Gödöllő HÉV, M4 to Bosnyák tér) are not listed: by Proposition XI.2 they leave every V_k unchanged.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0), `scripts/paper_augment.py`

the demand-weighted efficiency by 5.8%, against 1.3% for the greedy links. M5 is therefore primarily a capacity and accessibility project, not a redundancy project.

Pesti fonódó. The new tram connection Deák tér – Nyugati tér – Lehel tér joins lines 47/49 to lines 12/14 (Telex, 2024; BKK, 2025). It lowers the average vulnerability $\bar{V}$ by 13.1%, against 2.1% for the first two greedy links together, and raises the demand-weighted efficiency by 3.5%. At the same time it routes additional paths through the Lehel utca / Róbert Károly körút stop area, whose failure becomes more damaging (from 8.9% to 9.6%), so J as a whole does not improve. The Thököly út tram (Keleti – Bosnyák tér), which has been discussed but not planned, has only a small effect.

Layer	Project / next link	km	$\bar{V}$	$V_{\max}$	Worst failure	$-\Delta J$ (%)
Metro	<i>M5 inner section only</i>	—	0.052	0.320	Kálvin tér	27.8
	+1: Astoria – Corvin-negyed	1.21	0.050	0.181	Kálvin tér	55.1
	+2: Szent Gellért tér - Műegyetem – Déli pályaudvar	2.92	0.043	0.139	Corvin-negyed	64.7
	+3: Vörösmarty utca – Semmelweis Klinikák	2.88	0.039	0.112	Lehel tér	70.6
Tram	<i>Pesti fonódó only</i>	—	0.021	0.096	Lehel utca / Róbert Károly körút	−2.7
	+1: Vágány utca / Róbert Károly körút – Hun utca	0.47	0.021	0.087	Széll Kálmán tér M	5.3
	+2: Blaha Lujza tér M – Kálvin tér M	1.07	0.020	0.086	Széll Kálmán tér M	6.7
	+3: Budafoki út / Dombóvári út – Budafoki út / Szerémi sor	0.47	0.019	0.085	Széll Kálmán tér M	7.5

Table 17: Suggested next links once the planned project is built: greedy additions conditional on the project’s links. $-\Delta J$ is measured against the current network.

Source: own computation from the BKK GTFS Schedule feed (CC0-1.0), `scripts/paper_augment.py`

What to add next. Table 17 answers the planning question directly: if the planned project is built, which links should follow? The rows are the output of Algorithm 2 with the project’s links committed ($A_0 \neq \emptyset$). For the metro, the M5 is best complemented by Astoria – Corvin-negyed (1.2 km), which joins M2 and M5 to M3 without passing through Kálvin tér and lowers the worst case from 32.0% to 18.1%. It is followed by two longer outer links, Szent Gellért tér – Déli

pályaudvar (M4 to M2 on the Buda side, 2.9 km) and Vörösmarty utca – Semmelweis Klinikák (M1 to M3, 2.9 km). M5 with these three links reduces J by 70.6%, more than five greedy links on the current network (59.7%), and lowers Kemeny’s constant to 201.7 (Table 14). For the tram, the Pesti fonódó is best complemented by the short 1 – 12/14 link at Vágány utca – Hun utca, which removes the new weak point at Lehel utca / Róbert Károly körút, and then by Blaha Lujza tér – Kálvin tér and the Budafoki út gap. With these links the fonódó turns from a small loss (−2.7%) into a gain of 7.5% in J , while keeping its large gain in average accessibility.

XI.7. Interpretation and Limits

Three conclusions follow from the analysis. First, the Budapest metro is fragile at exactly two points, and the most effective remedy is a small set of bypass links that join the radial lines *without* passing through Deák Ferenc tér or Kálvin tér. Three links of about 1.3–1.8 km halve J . Second, extensions of existing lines, which account for most of the current metro plans, do not make the existing network more robust (Proposition XI.2). Third, planned projects that serve an existing hub (M5 at Kálvin tér, the Pesti fonódó at Lehel tér) improve average accessibility but shift the worst case to that hub. Each of them should be combined with the complementary links of Table 17.

The analysis is structural. The costs of new links are the median costs per metre of the mode, so construction feasibility, alignment and actual cost are not modelled, and the suggested links should be read as corridors (“connect these two lines between these two areas”), not as alignments. The station graph has no transfer penalty, so it overstates the value of links that create paths with an extra change of line. The demand weights come from the E1 prior, and the sensitivity analysis shows that the average-oriented recommendations depend on them. Finally, metro and tram are treated as separate layers. In a joint layer with walking transfers, a tram line can replace a failed metro station, so the joint vulnerabilities, and possibly the recommended links, would be different.

XII. Limitations and Future Work

We enumerate the principal limitations of the current model.

Idealised passenger behaviour. Independence assumption, single-leg transitions, and origin-constrained (not doubly-constrained) gravity model.

Time-varying supply. The time-homogeneous CTMC is valid only within a narrow service window where f_i^v is approximately constant.

Capacity constraints. The uncapacitated formulation ignores finite vehicle capacity. A capacity-dampened rate $\mu_i^v(N_i^v) = \mu_i^v \cdot \max(0, 1 - N_i^v / (K^v |N_{in}^v(i)|))$ is tractable for SSA but lacks a

closed-form stationary distribution.

Absence of validation data. No AFC or APC data were available. Parameters λ^v and γ are assumed, not estimated, and the prior uses the service-proxy estimator E1. Absolute passenger counts are therefore illustrative; the structural results (which stops dominate the ranking) are the robust part of the analysis.

Absorbing terminals. Stops without peak departures act as absorbing states and accumulate 19% of the passengers within 15 minutes (Section IX). An explicit trip-end (alighting) process would remove this artefact and allow longer horizons.

Mode coverage. Trolleybus and suburban rail are coded in the current feed with `route_type` values outside the framework’s mode table and are not part of the analysed network.

Markov assumption. Real service headways follow scheduled (deterministic) distributions, not exponential ones. Phase-type or semi-Markov extensions (Norris, 1997) can generalise this.

Single-mode assignment. Mode switching mid-journey is not modelled.

Per-mode resilience analysis. Node removal is evaluated within each modal chain, so the ranking captures the role of a stop within its own mode but not the loss of transfer opportunities between modes. A joint multi-modal generator $Q = \bigoplus_v Q^v$ with inter-modal transfer edges is required for that.

XII.1. Future Work

1. **AFC calibration:** Estimate λ^v from tap-in/tap-out distributions when farecard data are available.
2. **GTFS-RT integration:** Update f_i^v in real time from vehicle-position feeds.
3. **Doubly-constrained formulation:** Implement IPF balancing for trip distribution.
4. **Capacity-dampened SSA:** Nonlinear propensities with finite vehicle capacity.
5. **Joint and costed augmentation:** Repeat the link-augmentation analysis of Section XI on a joint metro–tram layer with walking transfers and transfer penalties, with construction costs per kilometre in place of a cardinality budget.
6. **Regional comparison:** Apply to Vienna (WL), Prague (DPP), and Warsaw (ZTM) GTFS feeds.
7. **Joint multi-modal criticality:** Node removal on the combined generator $Q = \bigoplus_v Q^v$ to recover cross-modal interchange significance.
8. **Trip-end process and full mode coverage:** Add alighting as an explicit outflow and include trolleybus and suburban rail.

XIII. Summary and Conclusions

We have developed a multi-modal, entropy-maximising Markov-chain passenger-flow model for the BKK Budapest GTFS network and implemented it in Python using sparse linear algebra throughout. The model operates on three complementary stochastic layers—a DTMC whose transition matrix is the Wilson softmax, a CTMC whose generator is derived from the same entropy kernel with GTFS-calibrated departure intensities, and an exact stochastic realisation via the Gillespie SSA.

Theoretical contributions. We derive the CTMC generator from entropy-maximising flows with correct dimensional analysis and sparse graph structure. We show precisely how the DTMC embedded jump chain relates to the CTMC, and characterise the conditions for ergodicity.

Empirical findings. On the BKK feed (5 177 stops, 6 450 scheduled links, four modes) the model conserves passengers to machine precision in the KFE and exactly in the integer SSA and τ -leap layers. The mean field is an accurate description at station level: exact SSA ensembles on the metro agree with it within 0.8% and fluctuate at the Poisson level $N^{-1/2}$, and τ -leaping on the full network agrees within 2.4%. Metro stations mix within a few headways, while terminal stops without departures absorb a growing share of passengers, which limits the time-homogeneous model to horizons of the order of a trip. In the resilience analysis the spectral gap turns out to be uninformative on every modal chain (the metro chain is periodic, the surface chains reducible), whereas Kemeny’s constant works as a fragmentation indicator. Criticality in the metro is concentrated in the three interchanges Deák Ferenc tér, Kálvin tér and Keleti pályaudvar, and only the inner cycle formed by M2, M3 and M4 offers redundancy against the loss of a station.

Network augmentation. Inverting the failure analysis gives concrete planning suggestions. The metro is fragile at two interchanges, and three links that join the radial lines around Deák Ferenc tér and Kálvin tér, without passing through them, halve the robustness functional; greedy search is within 5% of the exhaustive optimum. Line extensions provably add no redundancy, and planned projects that run through a hub (M5 at Kálvin tér, the Pesti fonódó at Lehel tér) improve average accessibility but move the worst case to that hub, so they should be complemented by the bypass links identified in Section XI.6.

Limitations. Without AFC or APC data, absolute passenger counts and λ^v parameters cannot be validated; the sensitivity sweep shows, however, that the default λ^v lies in a regime where the stationary concentration is insensitive to it. Capacity constraints, time-varying supply, trip ends and multi-leg journeys are absent from the current formulation.

Code and data availability. The full Python implementation is at <https://github.com/momarci13/bkk-markov-passenger-flow> (MIT licence). BKK GTFS: <https://opendata.bkk.hu> (CC0-1.0).

References

- Linda J. S. Allen. An introduction to stochastic epidemic models. In Fred Brauer, Pauline van den Driessche, and Jianhong Wu, editors, *Mathematical Epidemiology*, volume 1945 of *Lecture Notes in Mathematics*. Springer, 2008.
- Alex Anas. Discrete choice theory, information theory and the multinomial logit and gravity models. *Transportation Research Part B: Methodological*, 17(1):13–23, 1983.
- Beruházás.gov.hu. Gödöllői HÉV (H8) és csömöri HÉV (H9) vonalszakaszok korszerűsítése és az Örs vezér terén az M2 metróvonallal történő összekötése. <https://beruhazas.gov.hu/projektek/>, project no. 45998, 2026. Accessed 28 September 2026.
- BKK. Indulhat az új pesti fonódó tervezése. <https://bkk.hu/hirek/2025/09/indulhat-az-uj-pesti-fonodo-tervezese.15347/>, 2025. September 2025, accessed 28 September 2026.
- BKK. Tájékoztató a “Budafoki úti villamosvonal tervezése” projekt műszaki tartalmáról. Briefing to the Transport Committee of the Budapest General Assembly, 30 March 2026, 2026.
- Michiel C. J. Bliemer, Piet H. L. Bovy, and Bart Mevissen. Stochastic dynamic traffic assignment: A review. *Transportmetrica*, 2003.
- Yang Cao, Daniel T. Gillespie, and Linda R. Petzold. Efficient step size selection for the tau-leaping simulation method. *The Journal of Chemical Physics*, 124(4):044109, 2006.
- Oded Cats and Erik Jenelius. Dynamic vulnerability analysis of public transport networks: Mitigation effects of real-time information. *Networks and Spatial Economics*, 14(3–4):435–463, 2014.
- Robert B. Dial. A probabilistic multipath traffic assignment model which obviates path enumeration. *Transportation Research*, 5(2):83–111, 1971.
- Sven Erlander and Neil F. Stewart. *The Gravity Model in Transportation Analysis: Theory and Extensions*. VSP, Utrecht, 1990.
- Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977.
- Daniel T. Gillespie. Approximate accelerated stochastic simulation of chemically reacting systems. *The Journal of Chemical Physics*, 115(4):1716–1733, 2001.
- Bernd Heidergott, Geert Jan Olsder, and Jacob van der Woude. *Max Plus at Work: Modeling and Analysis of Synchronized Systems*. Princeton University Press, 2010.

- Dirk Helbing, Illés Farkas, and Tamás Vicsek. Simulating dynamical features of escape panic. *Nature*, 407:487–490, 2000.
- Jeffrey J. Hunter. The role of Kemeny’s constant in properties of Markov chains. *Communications in Statistics – Theory and Methods*, 43(7):1309–1321, 2014.
- Naunihal K. Jaiswal. Preemptive resume priority queue. *Operations Research*, 9(5):732–742, 1961.
- John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, 2nd edition, 1976.
- Vito Latora and Massimo Marchiori. Efficient behavior of small-world networks. *Physical Review Letters*, 87(19):198701, 2001.
- Mark Levene and George Loizou. Kemeny’s constant and the random surfer. *The American Mathematical Monthly*, 109(8):741–745, 2002.
- Sven Maerivoet and Bart De Moor. Cellular automata models of road traffic. *Physics Reports*, 419(1):1–64, 2005.
- Adam Meyerson and Brian Tagiku. Minimizing average shortest path distances via shortcut edge addition. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2009)*, volume 5687 of *Lecture Notes in Computer Science*, pages 272–285. Springer, 2009.
- George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14(1):265–294, 1978.
- James R. Norris. *Markov Chains*. Cambridge University Press, 1997.
- Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- Philine Schiewe and Marc Lindner. Direct-demand transit ridership models: A literature review of elasticity estimates. *Transportation Research Part A: Policy and Practice*, 165:103–121, 2022.
- Heinz Spiess and Michael Florian. Optimal strategies: A new assignment model for transit networks. *Transportation Research Part B: Methodological*, 23(2):83–102, 1989.
- Telex. Újbudától Újpestig egy villamossal – ez lehet a főváros leghosszabb villamosvonala. <https://telex.hu/belfold/2024/03/19/pesti-fonodo-budapest-ujpest-ujbuda-villamosvonal-2027-tervezes>, 2024. 19 March 2024, accessed 28 September 2026.

Telex. Hogy került hirtelen képbe pont a 3-as metró meghosszabbítása, és miért ilyen kurtán? <https://telex.hu/belfold/2026/07/22/3-as-metro-meghosszabbitas-rakospalota-ujpest>, 2026. 22 July 2026, accessed 28 September 2026.

Michael D. Vose. A linear algorithm for generating random numbers with a given distribution. *IEEE Transactions on Software Engineering*, 17(9):972–975, 1991.

Wikipedia contributors. Metro line M5 (Budapest metro). [https://en.wikipedia.org/wiki/Metro_Line_M5_\(Budapest_Metro\)](https://en.wikipedia.org/wiki/Metro_Line_M5_(Budapest_Metro)), 2026. Accessed 28 September 2026.

Darren J. Wilkinson. *Stochastic Modelling for Systems Biology*. Chapman and Hall/CRC, 3rd edition, 2018.

Alan G. Wilson. The use of entropy maximising models in the theory of trip distribution, mode split and route split. *Journal of Transport Economics and Policy*, 3(1):108–126, 1969.

Mehmet Yildirimoglu and Nikolas Geroliminis. Approximating the steady-state distributions of aggregated markovian traffic models. *Transportmetrica B: Transport Dynamics*, 6(4):319–340, 2018.

Jing Zhang, Feixiong Liao, Theo Arentze, and Harry Timmermans. A multimodal transport network model for advanced traveler information systems. *Procedia Computer Science*, 151: 244–251, 2019.

Appendices

A. Reproducibility

A.1. Software Versions

Package	Version
Python	3.11
NumPy	2.4
SciPy	1.17
pandas	3.0
NetworkX	3.6

Table 18: Software environment used for the results in this paper.
Source: own computation, *bkk/framework*

A.2. Reproducibility Checklist

1. `pip install -e .`
2. Download the feed from <https://opendata.bkk.hu> to `data/budapest_gtfs.zip`.
3. `python scripts/tdk_rerun.py --zip data/budapest_gtfs.zip`. This runs the network construction, generators, KFE, τ -leap and the node-removal sweep (`data/resilience_scores.csv`), then `scripts/paper_dynamics.py` (KFE trajectories, SSA and τ -leap ensembles, λ sweep; `data/paper_dynamics.json`) and `scripts/paper_tables.py`, which writes every table fragment in `paper/tables/`, and finally `scripts/paper_augment.py` (Section XI; `data/augment_results.json` and the `t_aug_*` tables, about 35 minutes). On Windows, `futtatas.ps1` performs all steps and builds the PDF.

Random streams are seeded (seed 42 for the pipeline, fixed seeds per realisation for the ensembles), so every number in Sections IX–X is reproducible.

A.3. Numerical Validation Tests

The package ships 91 `pytest` tests, all passing. Among them:

- Row sums of Q^v equal zero to $< 10^{-10}$ in ℓ^∞ norm.
- Row sums of P^v equal one to $< 10^{-14}$ in ℓ^∞ norm.
- $\|N^v(T)\|_1 = \|N^v(0)\|_1$ (KFE conservation) to $< 10^{-6}$ relative error.
- SSA: integer-valued state vector throughout; no negative entries.
- Prior: $\sum_i \hat{N}_i(0) = N_{\text{peak}}$ to $< 10^{-8}$ relative error.
- Metro platforms are aggregated to parent stations and the metro chain is irreducible with non-zero departure hazards.
- Eigenvalues keep sign and phase; reducible chains report an undefined Kemeny constant.
- The shortcut update (29) agrees with a full Dijkstra recomputation, with and without failures, on random graphs; pendant extensions leave every V_k unchanged (Proposition XI.2); links into k never reduce V_k (Proposition XI.3).

Symbol	Meaning	Unit / Range
n	Number of stops	5,177
m	Number of modes	4
R	Total admissible edges	6,450
C_{ij}^v	Generalised cost	s
λ^v	Entropy–cost multiplier	s^{-1}
$\pi_{j i}^v$	Wilson destination kernel	$[0, 1]$, $\sum_j = 1$
P_{ij}^v	DTMC transition	$[0, 1]$, row-stochastic
q_{ij}^v	CTMC off-diagonal rate	s^{-1}
μ_i^v	Per-passenger departure hazard	s^{-1}
$N_i^v(t)$	Passenger count	$\mathbb{Z}_{\geq 0}$
π^v	Stationary distribution	$[0, 1]^n$, $\sum = 1$
$K(P)$	Kemeny’s constant	steps
$g(P)$	Spectral gap	$[0, 1]$
$E(P)$	Network efficiency	dimensionless
I_i	Stationary departure exposure	pax s^{-1}
C_i	Stop criticality	$[0, 1]$
ω_{ij}	OD demand weight	$[0, 1]$, $\sum = 1$
c_{ab}	Cost of a new link $a-b$	s
$V_k(A)$	Vulnerability to failure of k after adding A	$[0, 1]$
$J(A)$	Robustness functional	$[0, 1]$

Table 19: Principal symbols and units.

Source: own computation, bkk/ framework

Notation Summary

B. Declaration on the Use of Artificial Intelligence

As required by the call for papers, I declare which artificial-intelligence tools were used in preparing this paper, and for what purpose.

Tool and version

Claude (Anthropic), Opus 5.5 model

Literature search and topic mapping

Yes, for the related-work overview in Section I.

Language checking and proofreading

No.

Code review and debugging

Yes, for the analysis code in bkk/ and scripts/.

Substantive text generation

No. The research question, the mathematical derivations, the implementation decisions and the interpretation of the results are my own work.

Where the depth of tool use in a given chapter or section differed from the above, a footnote

attached to the first sentence of that part says so.

C. Declaration of Originality

I declare that this paper is original: it is not identical to any paper of mine submitted to another section, to any paper submitted to another TDK conference, to a master's thesis submitted and defended, or to any published or forthcoming publication.

date

signature of the author